Matthew Emmanuel T. Labrador

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EDUCATION

University of the Philippines Manila *Manila, PH*
BS Computer Science — University Scholar | Expected May 2029
Current Cumulative GWA: 1.0375

PAREF Southridge School *Manila, PH*
Senior High School — Class Valedictorian, Gold Medalist | May 2025

EXPERIENCE AND PROJECTS

Tumbang Preso (TUMP) — DOST Gear Up NCR Esports Game Dev Challenge, 1st Place *2026*
Godot 4, GDScript, Blender, ENet Authoritative-Host Multiplayer, Dedicated Linux Servers

- Won 1st Place; now representing NCR at the National Finals in General Santos City, backed by DOST and partner companies.
- Sole developer, empty project to shipped build in five days: 3D models, characters, map, UI, sound design, bots, physics, netcode, and the full codebase.
- Engineered authoritative-host multiplayer over ENet, resolving contact by host-side distance after measuring 16 of 36 engine overlap callbacks silently failing.
- Deployed dedicated online lobbies on a Linux VPS with UDP discovery and join codes, reaching players behind carrier-grade NAT.
- Owned marketing and pitch, delivering the presentation and Q&A that won the regional.

eGovMed — eGov Hackathon PH, Champion (₱100,000 Grand Prize) *2026*
React, Vite, Node.js, Express, TypeScript, Redis, eGov API Suite, Vercel

- Won the grand prize with Bisaya Hackers for an AI triage system that assesses and routes patients before they queue.
- Developed full-stack across React and Node/Express, owning integration of eight national eGov APIs: triage, SSO identity, National ID eVerify, face liveness, messaging, payments, reporting, and chain anchoring.
- Gave every integration paired mock and live paths behind an environment switch, so it demos offline and refuses to boot in production while any service is mocked.
- Owned the pitch and presentation, delivering the live demo to the judging panel.

Diabetic Complication Swarm (GlycoSwarm AI) — AMD Developer Hackathon ACT II, Track 3 *2026*
Python, LangGraph, FastAPI, Next.js, Ollama, AMD MI300X GPU

- Lead developer of an international, cross-timezone team building a multi-agent diabetic complication screening system.
- Built it full-stack: a LangGraph StateGraph of four parallel agents each writing and running its own Python scoring code on NHANES lab data, a FastAPI service, and a Next.js front end.
- Served live inference with Gemma 4 26B on an AMD MI300X GPU via Ollama, with automatic failover to Fireworks GLM 5.2.
- Designed and delivered the live demo and pitch deck to the judges.

AI Application & LLM Development *2025–2026*
Python, Gemma 3B, Qwen 3, LangChain, LangGraph, Anthropic API

- Deployed open LLMs locally (Gemma 3B, Qwen 3) to study end-to-end setup, inference, and multi-agent performance.
- Built automated summarizer applications on the Anthropic API using RAG and Agent-to-Agent (A2A) architectures.
- Ran structured prompt evaluations across model versions and documented the findings.

UPLB Code Wars *2025–2026*
C/C++, Algorithms, Compiler-free Environment

- Solved complex algorithmic problems entirely in Notepad, with no compiler, internet, or AI assistance.

CHIP-8 Emulator *2026*
C/C++, Raylib, System Architecture

- Engineered a robust CHIP-8 emulator from scratch, implementing core CPU instructions and memory management.
- Built a custom visual debugger using Raylib to track system state, registers, and memory for analytical debugging.

Heart Disease Prediction Model *2025*
Python, pandas, Google Colab

- Trained a supervised learning model in Python to predict heart disease from patient features end-to-end.
- Cleaned data and performed feature engineering with pandas, handling missing values and categorical variables.

TECHNICAL SKILLS

Languages: Python, SQL, C, C++, TypeScript, GDScript, HTML
ML & Data: Feature Engineering, Evaluation Metrics (accuracy, precision, recall, F1), pandas, Prompt Engineering, Model Deployment
Agentic AI & Orchestration: LangChain, LangGraph, FastAPI, Multi-Agent Systems, Ollama
Methods & Tools: Statistical Analysis, Model Evaluation, Google Colab, Git, Linux/CLI, Raylib, Godot, Blender, Next.js, Vercel
Certifications: AMD Multiagent Systems Deployment, CS50 Intro to CS, AI Prompt Engineering, Python, C
Spoken Languages: Filipino (native), English, Latin

LEADERSHIP AND ACTIVITIES

Competitive Debater *2022–2025*
Various National Tournaments, Philippines
Quarterfinalist at Ateneo Peace Debate; competed at UP Diliman Debates, XSDC, and ASDC.

Google Developer Group and UP Socomsci *2025–2026*

Coordinated cross-functional teams on operations and event setup.

Math Team Captain *2022–2024*

PAREF Southridge School, Muntinlupa, PH

Led tutoring sessions and mentored peers for local math competitions.

AWARDS

1st Place, DOST Gear Up NCR Esports Game Dev Challenge (2026) — NCR representative to Nationals | Champion, eGov Hackathon PH (2026, ₱100,000) | DOST Undergraduate Scholar (2025–Present) | Top 5 Finalist, Olymphysics NCR (2025) | 5th Place, Philippine Statistics Quiz NCR (2025)